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DEVELOPMENT OF A ROBOTIC SYSTEM FOR PRESSURE VESSEL INSPECTION

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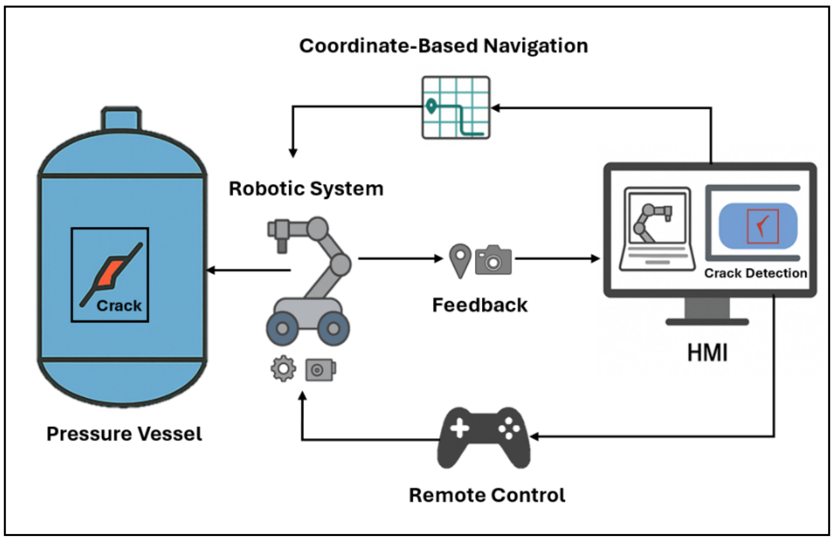
Graphical abstract



Abstract

The growing need for safer and more efficient inspection in industrial environments has driven the development of robotic solutions. This paper presents a fully integrated robotic system for inspection of pressure vessels, offering a low-cost alternative to complex Unmanned Aerial Vehicles (UAVs) and climbing robots. The system features a wheeled mobile base with a four-degree-of-freedom (DOF) manipulator, capable of extending, bending, and rotating the attached camera to access elevated and confined areas. Its dual-level control architecture includes an onboard Arduino-based controller for manual and autonomous motion, and a primary Windows Presentation Foundation (WPF) based Human-Machine Interface (HMI) for real-time monitoring, control, and AI-driven analysis. A custom-trained YOLOv11-Seg model, converted to Open Neural Network Exchange (ONNX) format for deployment, is embedded within the HMI to enable real-time crack detection and segmentation with 96% precision and 94% segmentation accuracy. Experimental validation demonstrated positional error of around 1% and system lag of 570ms. The novelty of this research lies in its all-in-one, end-to-end integration of mechanical design, dedicated software control, dual-level autonomy, and deep learning into a compact, deployable solution, removing the need for external lifting mechanisms or complex aerial stabilization, offering an intelligent robotic platform in industrial and broader safety-critical applications.

Keywords: Inspection, AI-Powered Detection, Robotics, Safety, HMI.

1.0 INTRODUCTION

Pressure vessels are critical components of power and chemical facilities, acting as containers for fluids or gases under high pressure in applications such as steam production, chemical processing, and power generation. These include boilers, storage tanks, heat exchangers, etc. They are made of durable materials such as steel or other alloys and come in various shapes and sizes [1]. Vessels operating under high pressure are subjected to significant stresses, making them vulnerable to damage resulting in catastrophic failures. Several pressure vessel failures have been reported in [2] that resulted in unwanted fatalities. An analysis of accidents related to pressure vessels reported in [3] finds the root cause of leaks, corrosion, weak joints, and overpressure. Hence, regular inspections are critical to ensuring structural integrity and safe functioning. Traditional inspection methods often involve human intervention, exposing them to contamination and other safety risks.

Pressure vessel inspection has gradually shifted from manual visual checks to robotic and automated systems. These modern methods improve both safety and efficiency. Considerable progress has been made with aerial platforms such as unmanned drones. They can reach confined or elevated areas without requiring human entry. Drones are often fitted with visual or thermal cameras to inspect hazardous zones. Magnetic crawler robots have also been developed for internal inspections. They use adhesion systems to move across metallic surfaces. Another approach involves programmable logic controller (PLC)–based robotic systems. These platforms combine vision modules and weld-tracking mechanisms to perform inspections autonomously. Many robotic systems now integrate non-destructive testing (NDT) methods. Examples include ultrasonic inspection, eddy current testing, radiographic imaging, and high-definition cameras. Such tools detect hidden flaws like corrosion, pitting, or cracks. Drones are increasingly common for external surveys and difficult-to-reach internal areas. They are frequently equipped with thermal cameras or LiDAR for 3D mapping. In recent years, computer vision and machine learning have been applied to defect detection. These methods analyze collected images or video automatically, reducing reliance on human judgment and improving consistency.

Although there has been extensive research in robotics for inspection within industrial environments, much focus has concentrated on designing UAVs and climbing robots to inscribe inclined or vertical surfaces. Certain pressure vessels, as extremely large, tend to necessitate inspection at considerable heights, which can easily be accomplished through them because of their high flexibility within such environments. But even with these developments, most solutions are still dependent on sophisticated control systems, subtle stabilization processes, and have a hard time with non-regular geometry or need expensive hardware. Climbing robots tend to be cumbersome because of the necessity of sophisticated gripping systems (e.g., suction, magnets, mechanical grippers) and high-level control algorithms to move across surfaces [4]. Similarly, UAV-based inspection systems require intricate control strategies to stabilize flight. They struggle with limited flight time, GPS dependence, high cost, and indoor applicability [5]. Apart from mechanical design, progress in metallic fault detection still faces many challenges. Datasets are often too small. Class imbalance is another problem, and performance also varies across different materials and defect types. High processing power is required, which makes real-time use in industry difficult. In addition, very few studies combine mechanical design, instrumentation, and AI-based defect detection into a single framework.

To address these challenges, this work presents a simple but distinct solution. The study features a wheeled robot built for inspection tasks. The robot carries a manipulator with four degrees of freedom (DOF). A dual-level control system has been incorporated, along with a vision-based AI model for flaw detection. Since the robot moves on the ground, there is no need for flying mechanisms. This reduces both energy consumption and software complexity. The design allows inspection of pressure vessels without complex control systems or heavy fabrication processes. Despite this simplicity, the system remains flexible in operation. The robot can travel smoothly on the ground. Its manipulator extends to different heights, enabling a mounted camera to reach high or narrow inspection points that would otherwise require more advanced systems. The manipulator is designed with a bending function that enhances its ability to reach confined areas, allowing for more detailed inspections. The robot can also be operated remotely and is capable of coordinated motion, combining vertical extension with rotational scanning. This removes the need for external lifting devices or stabilization mechanisms. Control of the system is based on a hierarchical structure, consisting of a primary controller and a secondary Arduino-based controller. The primary controller is developed as a Windows Presentation Foundation (WPF) application, which serves as the Human-Machine Interface (HMI) for issuing commands, collecting data, and monitoring operations in real time. A further distinction of this work lies in the integration of a custom HMI with a crack detection and segmentation module, based on the YOLOv11 framework. This creates an end-to-end solution that brings together mechanical design, control architecture, and defect identification. The embedded vision module allows automatic evaluation of metallic surfaces, making the inspection process both faster and more reliable. The final model was converted to Open Neural Network Exchange (ONNX) format and integrated into the HMI to allow real-time crack detection on live or recorded video once a camera is mounted on the robot. The deep learning model demonstrated superior detection and segmentation accuracy, outperforming existing models like Mask R-CNN [6] and ADDU-Net [7] in both speed and precision. The novelty of this study is based on an all-in-one, end-to-end approach covering the mechanical design, control architecture, AI-powered detection, and overall system integration and delivering a complete inspection solution through a simplified approach across all domains, including control logic and hardware design.

The remainder of this paper is as follows: Section II reviews the related work and highlights current trends in inspection technologies. Section III details the materials and methods involved in the development of the system, including the mechanical construction of the prototype and techniques of control system design. Section IV explains the design of the AI-based crack detection system and its integration into the Human-Machine Interface (HMI). Section V presents results and discussions and provides testing results of the prototype. Section VI includes a conclusion by summarizing the key findings of the paper.

2.0 RELATED WORK

Robots for industrial maintenance minimize human involvement and enable inspections in hazardous or inaccessible areas. Several review studies have highlighted different types of robots and their research trends and challenges in inspection robotics across various industrial applications. For instance, [8] discusses the importance of robots and aerial vehicles for inspections in industrial facilities, emphasizing aerial vehicles and robots for confined areas and elevations. A systematic review by [9] on the development of robotic manipulators focused on Modelling, control, adaptability in harsh environments, and the complexity of hyper-redundant designs. Wheeled-type inspection robots were reviewed in [10] for pipeline inspections, exploring their designs for navigating inside varying geometric areas and highlighting their limitations in vertical movement. Climbing robots for inspection have been widely studied, particularly in terms of their locomotion mechanisms and control strategies; however, researchers like [11] have pointed out their inherent challenges, especially in maintaining reliable adhesion and control complexity. Robotics and autonomous systems have also been explored for use in hazardous environments such as inspection and disaster response, where autonomy and adaptability remain significant technical hurdles [12]. Additionally, several focused studies have contributed to developing robotic systems addressing different inspection challenges in various industrial facilities. For example, [13] developed the Robotic Boiler Header Inspection Device (BHIR) for inspecting horizontal boiler header pipes. BHIR-I aids manual inspections with a borescope, while BHIR-II is autonomous with CCD (charged-coupled device) cameras and a redundant localization system. For structural inspection,[14] proposed a multi-legged inspection robot for steel structure, featuring a novel twisting motion for crouching and a fine adjustment mechanism using a linear actuator. The PETROBOT project [15] comprises three robotic systems for petrochemical pressure vessel inspections, using cameras, ultrasonic probes, eddy current sensors, and 3D mapping for anomaly detection. Enhancing mobility,[16] presented a crawler mobile robot with sub-crawlers utilizing compliance control in rotary joints for enhanced ground adaptability. The robot integrates semi-autonomous control, allowing it to traverse obstacles without prior environment sensing. A modular platform Straddle Root Inspection Robot, a 2-DOF Crawler Robot, and an ESP Gas Detection Robot was developed by [17], a robotic platform with multiple configurations utilizing multi-level control for navigation, sensor fusion, and tasks. Meanwhile, [18] designed WATER-7, a pneumatically driven inchworm-like Robot for autonomous pipeline inspection equipped with thrust and active bending modules for high maneuverability. [19] explored swarm intelligent systems for jet turbine inspections to optimize robot control policies. While the macroscopic model and simulations were well-detailed, there was limited mention of extensive real-world validation with physical robots. A deep learning-powered spherical tank inspection robot that leverages weld line recognition and Eulerian circuit-based path planning was later introduced [20], combining AI and robotics to enhance inspection efficiency. EspeleoRobo, a semi-autonomous platform [21], was designed for confined space inspections using Simultaneous Localization and Mapping (SLAM) and vector-field navigation. [22] designed a manipulator-based inspection system with a 3D vision for object detection, but requiring setup on-site, increasing downtime and complexity. Another SLAM-based research was presented in [23], a smart mobile robotic system equipped with multimodal sensors, SLAM navigation, and A* path planning, but was constrained by the lack of vertical elevation capability. In contrast, [24] introduced a Programmable Logic Controlled (PLC) pressure vessel inspection robot using Mitsubishi FX5U-32MT and FX5-40SSC-S controllers, along with magnetic tracks enabling stable weld seam tracking on curved surfaces but limiting height access to the center of the structure. An advanced robot called CRAS is another benchmark in inspection robotics, which uses magnetic adhesion and ultrasonic sensors to inspect welds at high temperatures, improving safety [25]. [26] introduced the AIR Robot, which transitions between multiple autonomy levels from manual to fully autonomous control. A PLC-controlled tractor-trailer robot was designed by [27] with a vision system for weld line tracking and better adaptability on curved surfaces. Aerial vehicles are increasingly utilized for inspection procedures, such as [28], which addressed the challenge of stabilizing a wall-climbing robot (HORNET) under crosswind conditions. Their robot, equipped with propellers for lift and a gyro sensor for posture detection, utilized a PD control system to stabilize the yaw angle and resist lateral drift caused by wind. Studies by [29] highlighted the use of Aerial Robots using model predictive control (MPC) and visual-inertial (VI) motion estimation for inspection in confined and GPS-denied environments, specifically targeting thermal power boiler systems. Similarly, [30] introduced an Unmanned Aerial Vehicle (UAV) based remote-controlled inspection system integrating stereo cameras and VI navigation for 3D mapping and obstacle Avoidance. Likewise, [31] presents automated mission planning for UAV-based inspections in large-scale power plants using Computational Fluid Dynamics (CFD) based turbulence mapping to optimize routes and ensure operational safety. Beyond mechanical inspection, modern maintenance demands integrated nondestructive testing (NDT) methods, particularly for detecting surface flaws like cracks. Recent developments in deep learning have transformed crack detection into a task of pixel-precise classification and instance segmentation. For instance, [32] pioneered a two-stage process for bridge decks, first detecting cracks with YOLOv7 (outperforming Faster R CNN and RetinaNet) and then refining their U Net segmentation to measure crack length and width with 92.4% accuracy, although only two bridges were evaluated, potentially limiting generality. [33] demonstrated that, even with just twenty training images, a clever two-stage data augmentation strategy, especially rotation, can push recall up to 96%, yet reliance on such a tiny dataset might be a risk. Meanwhile, [34] showed that preprocessing concrete images into grayscale before feeding 40,000 RGB frames into a VGG16-based CNN yielded F1 scores above 99% for crack classification, but they stopped short of pixel-level segmentation, leaving precise localization for future work. Further, ADDU-Net [7] introduced asymmetric decoders and dual attention to segment multi-scale cracks robustly under shadows/debris. Extending the frontier of object detection, [35] introduced YOLOv11 with architectural enhancements like C3k2 and C2PSA, significantly improving performance across tasks, and achieved up to 54.5% mAP50–95 at 13ms latency, outperforming earlier YOLO variants in both speed and accuracy. Similarly, [36] presented a refined YOLOv5s-based model for steel surface defect detection, integrating multiscale feature extraction (MSFE) and efficient feature fusion (EFF) modules, which outperformed existing models on the NEU-DET dataset with fewer parameters. Complementing these detection advances, the work by [6] on implementing Mask R-CNN in CVAT via AWS EC2 showed a practical solution for automated annotation workflows, though it bears the trade-off of high dependency on cloud infrastructure that may cause scalability issues.

The literature highlights a wide range of robotic systems developed for industrial inspection. These include wheeled robots, climbing robots, multi-legged platforms, and unmanned aerial vehicles (UAVs). Each category is designed to address particular challenges such as navigating narrow passages, reaching vertical points, or ensuring stability in harsh environments. Climbing robots and UAVs are effective for accessing elevated or restricted areas, but they often face limitations related to design complexity, control stability, endurance, and cost. Wheeled robots provide reliable movement along the ground, though they usually lack mechanisms for vertical extension. At the same time, progress in automated defect detection has transformed crack analysis from simple identification into advanced tasks of classification and segmentation. Methods like YOLO, Mask R-CNN, and ADDU-Net have contributed significantly to improving both accuracy and processing speed. Building on these gaps, the present work proposes a wheeled robot that not only ensures ground mobility but also extends its inspection reach, offering a more balanced and practical solution. It has been equipped with a manipulator to overcome vertical reach limits, ensuring stability, simplicity, and cost-effectiveness, as well as accessibility in limited areas. Unlike most previous research, which treated mechanical design, control systems, and AI-based flaw detection as separate areas, this study presents an integrated solution that combines all these elements into one cohesive system.

3.0 METHODOLOGY

3.1 Mechanical Design

The inspection robot has a circular chassis and a 4-DOF manipulator in the center. This manipulator can hold either a camera or a non-destructive testing device. The chassis is 18 inches in diameter and is supported by two NEMA-17 stepper motors and four caster wheels. This setup gives the robot stability and lets it move in the XY direction on the ground. The chassis is made from Hylam fiber because it is lightweight and durable. The manipulator, centrally mounted on the chassis, enables advanced mobility for surface inspection. Its first DOF provides a 0–90° angular motion, allowing the manipulator to transition between a compact homing position for navigating areas with limited clearance and an upright working position. The second DOF utilizes a 30.5cm (12 inches) rack and pinion mechanism, powered by an MG996R servo motor, to achieve Z-axis mobility, enabling the manipulator to elevate at different heights. The manipulator can extend to a height of 73.66cm (29 inches) at full reach. The third DOF, mounted on the rack, provides 360° rotational motion through another MG996R servo motor, enabling all-inclusive coverage of the inspection area. The fourth DOF, mounted on top of the third DOF, features a 4-bar mechanism powered by a N20 micro-geared DC Motor that carries the inspection equipment (i.e., camera) and offers reciprocating motion. Together, they provide combined rotational and reciprocating mobility for precise inspections. All the actuators used in the prototype were selected based on predetermined torque requirements according to the mass of the robot. The motors used for the manipulator are relatively lightweight compared to the stepper motors used for the wheels, ensuring the manipulator remains stable.

Figures 1 and 2 depict the Computer-Aided Design (CAD) model of the prototype and the motion capabilities of the robot. Fig. 1 highlights the manipulator in both the extended and homing positions. The top-left image in the collage illustrates the design of the 3rd and 4th DOFs, where the servo motor rotates the four-bar mechanism mounted on top of it. Fig. 2 outlines the various mobility features of the robot and explains their movement.

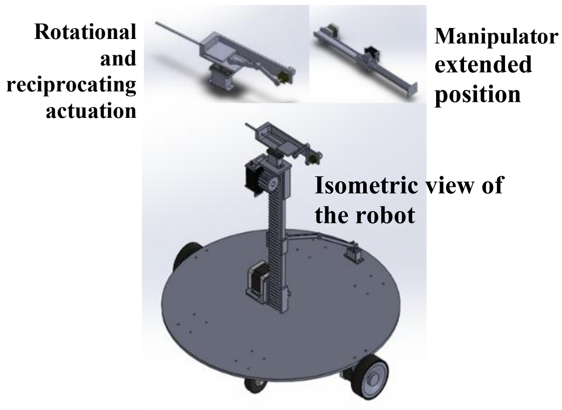


Figure 1 Inspection Robot: 3D Model.

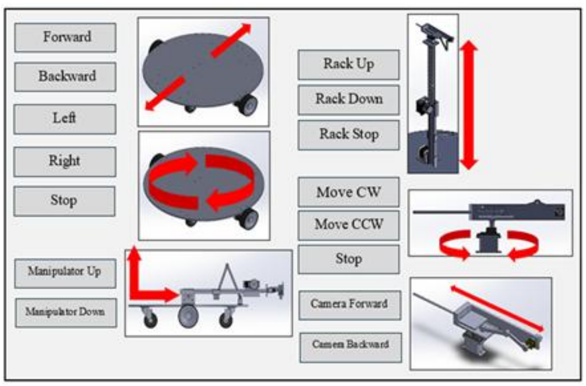


Figure 2 Inspection Robot: Mobility Features

Figure 3 shows the physical prototype. The stepper motors at the wheels are equipped with encoders, and the servo motor powering the gear is also equipped with a sensor to track its motion.

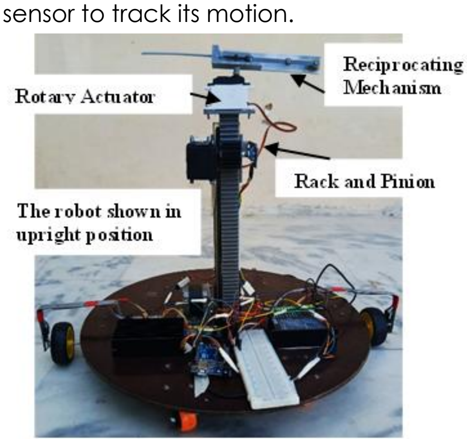


Figure 1 Inspection Robot: Physical Prototype

3.2 Control System

The robot's control system consists of two levels: the primary control system is operated remotely by an operator using dedicated software designed for the robot. This software handles command lines, data acquisition, and monitoring. The secondary control system is managed through an Arduino Mega 2560, which controls the onboard actuators and sensors (see Fig. 4).

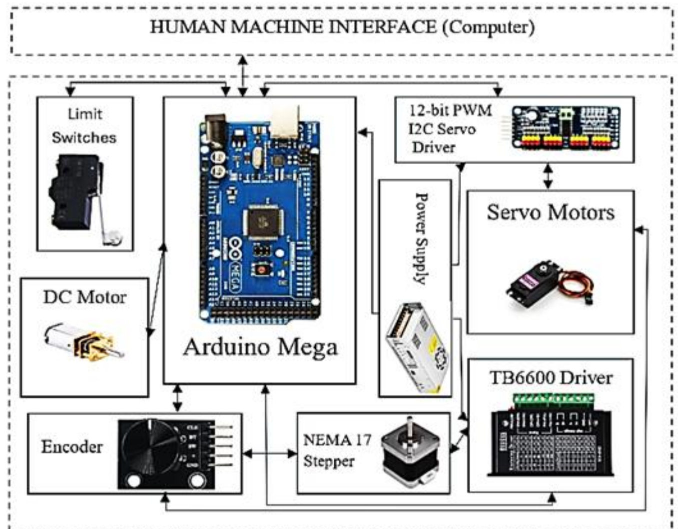


Figure 4 Block Diagram depicting Primary and Secondary Control Integration

The Arduino receives commands from the upper controller and directly controls the actuators while acquiring sensor data. The secondary control system is divided into two levels: manual control, where the operator controls all the robot's movements, and autonomous control, where the robot is provided with predefined XYZ coordinates for the inspection location and other required motions. The robot then sequentially calculates and executes the line of commands. The system initiates by asking for the COM port position and baud rate to establish communication between the HMI and Arduino. These parameters are selected as defined in the Arduino code. Once the connection is detected, the status changes to "Arduino Connected" at the bottom left of the screen, as shown in Figures 5 and 6.

The system includes a manual control interface with specialized buttons, as shown in Figure 5, for managing the chassis and manipulator, allowing the user to send out instructions such as XY motion of the base wheels, up/down manipulator movement, clockwise and anticlockwise movement of a rotary mechanism, and 4-bar mechanism motion. These commands are transmitted via PC for real-time control. The robot is equipped with KY-040 rotary encoders installed on each wheel and the manipulator, which tracks the robot's position.

Figure 2 User Interface: Manual Control

The autonomous control system operates on pre-calculated inspection points within a pressure vessel, focusing on stress concentration areas such as welded joints and nozzles. The coordinates of joints relative to a reference position are pre-determined. In this mode, the robot operates using a structured process that begins with selecting all the required actions sequentially that the robot needs to perform, along with their corresponding values, which can either be numeric (such as XY coordinates or elevation heights) or strings representing specific commands. Once the desired sequence of actions is defined, the HMI converts these instructions into a structured CSV format, which outlines each command line for execution. These command lines are displayed in a tabular format within a DataGrid interface as depicted in Figure 6, allowing the operator to review and verify the task list before initiating execution. The "Open" button allows the user to select a pre-configured file that has already been set according to the location of the inspection point. The "Read" button enables the software to read all the commands, and the "Send" button sends these commands to the Arduino for execution. Users have the option to import a pre-configured CSV file, which is automatically loaded into the same DataGrid for review. Each task is listed sequentially with its corresponding status (e.g., pending or executed. Once started, the HMI interface dynamically updates the status of each task from pending indicating that the required action has not been completed to executed in real time, providing continuous feedback on the robot's progress. At the secondary control level, which is Arduino-based, the command line is processed using parsing, which is the process of analyzing a string of text or data to extract commands by breaking it into separate components that a program can access. Each component corresponds to a specific motor action and contains the necessary values and strings for executing the task. These commands are then executed sequentially by the robot to reach and inspect the desired point. Figure 6 shows the user interface for the data grid listing sequential commands for autonomous control, containing identifiers and their corresponding action values. For instance, in ground-level XY motion, the identifier is "X-Y Motion," the numeric values correspond to the X- and Y-coordinates.

For rack actuation control, which involves only Z-axis movement or elevation, only one value is provided, corresponding to the height it needs to move to. The reciprocal actuation control takes a string command since it only moves either forward or backward, with its motion constrained by limit switches. This method applies to all motion types, and the robot performs each task sequentially based on the provided parameters.

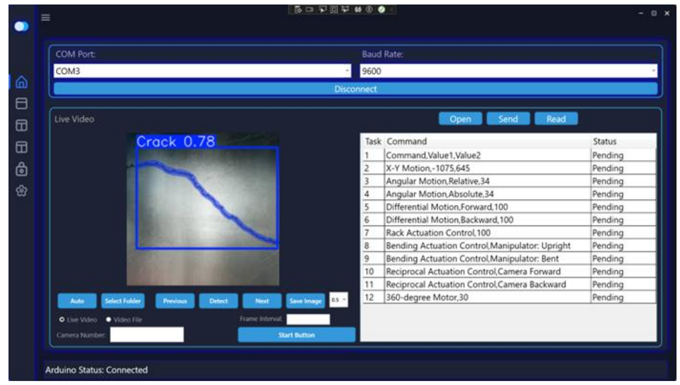


Figure 3 Image Detection, Communication, and user interface for the data grid listing sequential commands.

3.3 Ground Navigation Control

After the robot gets its location to inspect, the navigation system incorporates several key calculations that involve trigonometric principles to travel to the point. This involves calculating the angle and distance towards the location. The angle required for the robot to rotate towards a target point is determined using Eq. (1).

$$\boldsymbol{Required\ Angle} = \boldsymbol{tan^{-1}}\left(\frac{\triangle\, \boldsymbol{y}}{\triangle\, \boldsymbol{x}}\right) \times \frac{\mathbf{180}}{\boldsymbol{\pi}} \tag{1}$$

Here, Δx and Δy represent the differences between the robot's current coordinates and the new coordinates provided. The straight-line distance to the target, or hypotenuse, is calculated using Eq. (2), the Pythagorean theorem.

$$\boldsymbol{Distance} = \sqrt{(\triangle\, \boldsymbol{x})^{\mathbf{2}} + (\triangle\, \boldsymbol{y})^{\mathbf{2}}} \tag{2}$$

For turning, the arc length required for the wheels to rotate the robot by the desired angle is computed using Eq. (3).

$$\boldsymbol{Turning\ Distance} = \frac{\boldsymbol{\pi} \times \boldsymbol{wheel\ base} \times |\boldsymbol{angleDifference}|}{\mathbf{360}} \tag{3}$$

The wheelbase is the distance between the two wheels. The angle difference represents the angular displacement required for the robot to align with the goal direction. This turning distance is then converted into motor steps in Eq. (4).

$$\boldsymbol{Steps} = \frac{\boldsymbol{Turning\ Distane}}{\boldsymbol{\pi} \times \boldsymbol{wheel\ diameter}} \times \boldsymbol{steps\ per\ revolution}\ (\mathbf{6400}) \tag{4}$$

Similarly, the straight-line distance for forward or backward motion is converted into steps in Eq. (5).

$$Steps = \frac{Distance(mm)}{\pi \times wheel\ diameter} \times steps\ per\ revolution\ (6400) \quad (5)$$

The TB6600 stepper driver was used to operate steppers, and it is configured for 32 micro-steps per full step, enhancing the stepper motor's resolution. For a standard stepper motor with 1.8° per step, this configuration increases the steps per revolution from 200 to 6400, enabling smoother and more precise movements.

4.0 Visual Flaw Detection Through HMI-Integrated AI System

For the inspection of pressure vessels, a deep learning-based crack detection module was developed and integrated into the primary control layer of the robot, the Human-Machine Interface (HMI). As the robot autonomously navigates within the vessel and reaches key stress-concentration zones such as weld joints, nozzles, and curved surfaces, the mounted inspection camera captures high-resolution images or live video streams of the metallic surfaces. These visual inputs are sent directly to the HMI, where an embedded AI model analyzes them for structural flaws, enabling real-time defect detection.

4.1 Dataset Preparation and Model Training

The model was trained using a custom-built dataset. This dataset was compiled by aggregating small open-source image sets of cracks occurring in metal sheets, pipelines, welded seams, and machined parts. Each image was carefully labeled using the Computer Vision Annotation Tool (CVAT), with both bounding boxes and segmentation masks. To enhance the dataset's robustness and improve the model's generalization across various lighting and texture conditions, augmentation techniques were applied, including random rotation, flipping, translation, brightness adjustment, contrast modification, and noise injection. The final dataset comprised 1,357 images, split into 1,111 for training and 246 for testing.

Training was conducted using Google Colab with an NVIDIA Tesla T4 GPU. The model used was the YOLOv11-segmentation model, and a supervised learning approach was employed. The model architecture introduced lightweight convolutional blocks to reduce latency while improving object recognition accuracy, particularly for small or partially occluded cracks. It was trained for 200 epochs with a batch size of 16 and a 640×640 input resolution. The initial learning rate was 0.002 and Automatic Mixed Precision (AMP) was enabled to accelerate training without loss of precision. Early stopping was applied after epoch 160 due to a plateau in performance. Post-training, the model was converted into ONNX format for cross-platform compatibility and deployment within the HMI.

4.2 HMI Integration and Inspection Interface

The trained model was embedded into a WPF-based Human-Machine Interface, offering tools for image and video-based inspection. Once the robot begins inspection, the camera feed is routed to the HMI, where the operator can choose between manual and automatic modes for crack detection.

4.2.1 Image Inspection

On the HMI's Home Page (Figure 6), users can select an entire folder containing captured inspection images. The interface provides navigation tools to scroll through images individually using Next and Previous buttons. A Detect button applies the crack detection algorithm to the currently viewed image, and the result can be saved using the Save button. A threshold adjustment ComboBox allows users to control the model's detection sensitivity, making the system adaptable to different lighting and surface conditions. Users can enable automated batch processing by pressing the Auto button. The system sequentially loads all images from the selected folder, runs the detection algorithm on each one, and stores the processed results, including bounding boxes and segmentation masks, in a new folder automatically generated on the desktop. This feature significantly reduces manual effort and accelerates batch analysis workflows.

4.2.2 Video Inspection

The system also supports crack detection from both pre-recorded videos and live camera feeds, as depicted in Figure 7. Users can select a video file through the HMI and set a frame interval to reduce computational overhead. The system inspects frames at the defined interval and compiles the processed video into an output file named Output_Filename.mp4, saved in the Video Detection folder on the desktop. For real-time inspection, users can connect to a camera by entering the camera index (e.g., 0 for the default webcam). On the left side of the interface, the "Live Video" feed from the inspection camera shows the detection of a surface crack, with a yellow bounding box and a confidence score of 0.81. It also provides control options for selecting folders, detecting cracks, navigating between frames, and running video with a specific frame interval. On the right, the processed video window highlights the detected crack with a blue bounding box and segmentation mask, displaying a confidence score of 0.83. Pressing the Start button initiates live analysis. The system processes the video feed frame-by-frame, marking detected flaws, and stores the output in the same Video Detection folder. This feature enables on-site, real-time defect identification during robotic deployment.

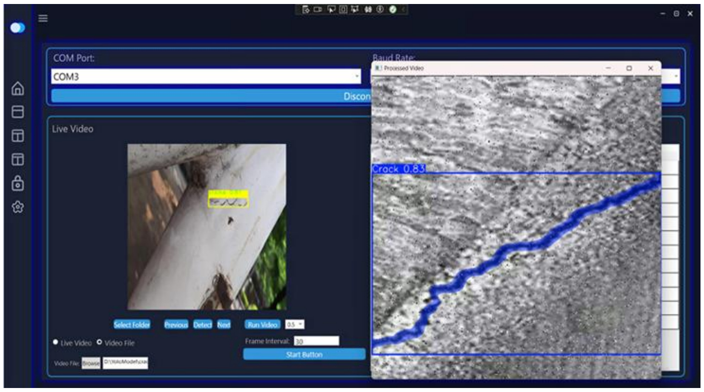


Figure 4 Real-Time Video-Based Crack Detection and Segmentation Interface

5.0 RESULTS

The robot was tested in the lab environment using a set of predefined coordinates to assess motion capabilities. Performance evaluation was conducted through multiple tests, including movement accuracy and response time. Encoder readings recorded positional and movement data, while timestamps were logged for measuring response time. The first set of tests was conducted to measure the response time of the robot. The calculated response time was determined to be 530ms, while the actual response time measured was 1100ms, resulting in a lag of 570ms, which is the delay between command transmission and its execution by the robot. Each set of commands for mobility tests and lag response measurements was executed and tested several times to ensure the data was reliable. Each mobility test was conducted individually to evaluate the time lag for each task. The theoretical response time was calculated by measuring the response times of components such as the sensors, actuators, and the controller. The observed lag resulted from several factors. One factor was communication latency, which came from delays in data transmission between the control system and the robot. Processing delays caused by the onboard controller also contributed. Mechanical issues, like actuator inertia and friction, slowed the robot's physical response. Additionally, a significant source of the delay was the continuous monitoring of data by the encoders. This further contributed to the overall lag and created a bottleneck, limiting the speed of data processing based on the encoders' feedback, which slowed the system's performance. The second set of tests was conducted to evaluate positional accuracy. The primary challenge for the robot is to navigate to the correct inspection location. The robot is provided with the coordinates and then calculates the distance it needs to travel to reach the location using Eq. (2), as described in the ground navigation control section. A set of XY coordinates was provided to the robot, and the distances were calculated. These values were compared with the actual distances travelled by the robot to assess its accuracy. Percentage errors were then calculated using Eq. (6). Table 1 presents experimental data, showing the robot's movement to several commanded locations. The displacement errors were due to the following reasons. Micro-stepping nonlinearity in stepper motors, micro steps are not perfectly linear; actual displacement per micro step can vary and reduce accuracy, causing minor deviations. Another factor was the encoder noise, causing minor feedback effects. Additionally, swivel misalignment in caster wheels and friction between the wheels also deviates movement precision. Collectively, these factors contributed to the deviation between the theoretical and actual responses.

$$\%\,Error = \left(\frac{|Actual\ Distance - Calculated\ Distance|}{Calculated\ Distance}\right) \times 100 \qquad (6)$$

Table 1 Performance Analysis: Ground Navigation Accuracy

X (cm)	Y (cm)	Calculated Distance (cm)	Actual Distance (cm)	Error (%)
20	35	40.311	41	1.7
38	57	68.505	69.5	1.4
68	45	81.541	82.5	1.2
89	70	113.229	115	1.6

In parallel, the crack detection model integrated into the HMI was evaluated using the YOLOv11-seg architecture. The model demonstrated robust performance on the test dataset, achieving a precision of 96.58% and a recall of 93.43% for bounding box detection, while segmentation precision and recall reached 94.69% and 91.61%, respectively. At an IoU threshold of 50%, the model achieved mAP50 scores of 96.88% (detection) and 93.53% (segmentation). It accurately detected and segmented a wide range of crack types, both thin and thick, on materials such as metal sheets, pipelines, and welded joints, with confidence scores ranging from 0.8 to 0.9 as depicted in figure 8.

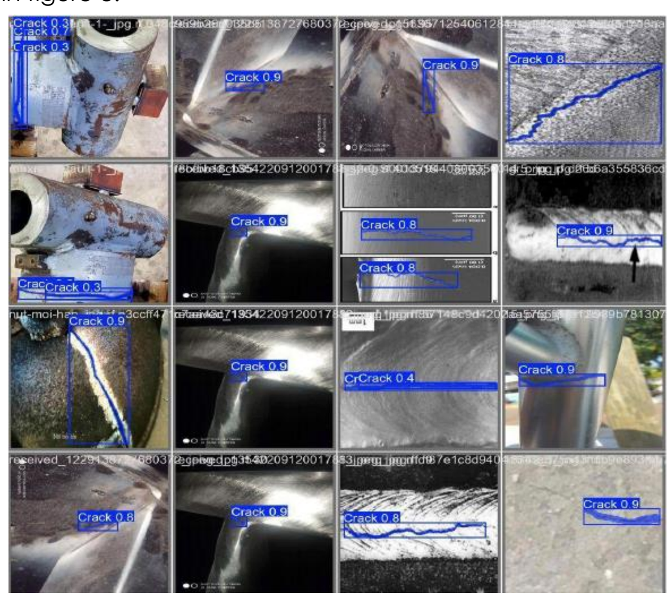


Figure 8 Detection and Segmentation Results on Diverse Surface Crack Images

The confusion matrix, as depicted in Figure 9, demonstrates the performance of the model in distinguishing between cracks and background regions. The matrix shows that the model successfully detected cracks in 265 instances but missed cracks in 42 cases where cracks were present but not identified. Moreover, 9 instances were incorrectly predicted as cracks where no cracks existed.

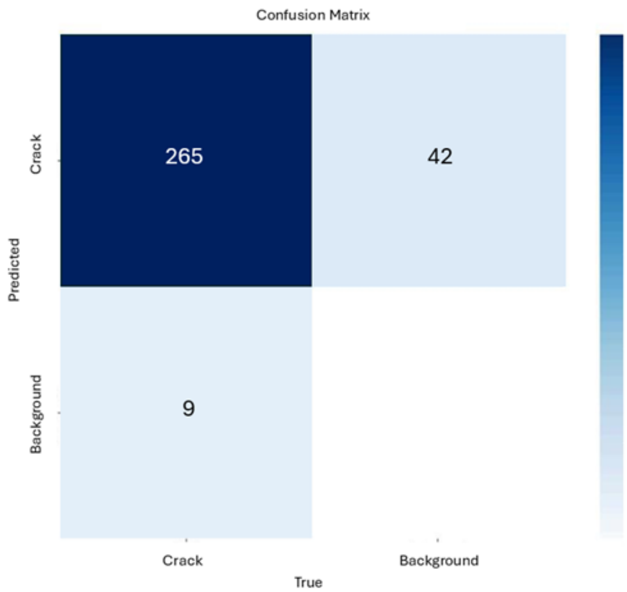


Figure 9 Confusion Matrix Showing Crack Detection Performance

Despite occasional low-confidence results, attributed to dataset limitations, the model processed images at 12.71 milliseconds per frame (or ~37 FPS), making it ideal for real-time inspection scenarios. Compared against other leading models like U-Net [32], ADDU-Net [37], KTCAM-Net [7], Efficient U-Net [38], and Mask R-CNN [39], YOLOv11 consistently outperformed in all key metrics, confirming its reliability and efficiency.

6.0 CONCLUSION

This work demonstrates the functionality and performance of an autonomous inspection robot designed specifically for pressure vessel inspection, integrating a dual-level navigation system with AI-powered visual flaw detection capabilities. The robot was validated through lab experiments, where onboard encoders enabled path tracking and motion estimation. The control architecture, based on dual autonomy modes, offered flexible operation, and the system achieved a small positional error of around 1% and a lag of 570 ms only, confirming its suitability for precise, real-time inspection tasks. While the current prototype serves as a successful proof of concept, future iterations could improve responsiveness by incorporating more efficient actuators and replacing the Arduino Mega 2560 with a higher-performance controller to reduce processing delays and mechanical inertia. A notable contribution of this work is the integration of a deep learning-based crack detection system into the HMI using the YOLOv11-seg model, which achieved 96.58% detection precision and 94.69% segmentation precision, with an average inference time of 12.71 ms per image. The system enables both live and recorded image and video inspection through a WPF interface, offering threshold adjustment, automated batch processing, and real-time defect visualization. Compared with existing models such as Mask R-CNN, KTCAM-Net, and ADDU-Net, the proposed system demonstrated superior performance in both accuracy and speed, making it a viable industrial solution for rapid, automated inspection of complex metallic structures. Although the model showed occasional limitations in detecting very fine cracks or under low-light conditions, these can be addressed in future work by expanding the training dataset and enhancing the model's adaptability. The results underscore the importance of combining deep learning with HMI tools, providing a scalable solution not only for pressure vessels but also for broader applications in the aerospace, construction, and manufacturing sectors, where reliable structural health monitoring is critical. While the prototype demonstrates efficient operation, it primarily serves as a proof of concept for this type of design. Future iterations could incorporate advanced sensors and actuators to enhance performance, adaptability, and reliability in industrial environments.

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